

Diag Solutions

1. **1.** If $\cos \theta = 0$, then the value of θ (where $0^\circ \leq \theta \leq 90^\circ$) is:

Answer: (C)

Solution:

1. Look for the angle where cosine is zero

Derivation:

$$\begin{aligned}\cos 90^\circ &= 0 \\ \implies \theta &= 90^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

2. **2.** The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is:

Answer: (A)

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$

Derivation:

$$\sin^2 \theta + \cos^2 \theta = 1 \text{ for any } \theta$$

Introduction to Trigonometry — Trigonometric Identities

3. **3.** If $\sec \theta = 2$, then θ is:

Answer: (B)

Solution:

1. Recall that $\sec \theta = 1/\cos \theta$
2. Identify the angle where $\cos \theta = 1/2$

Derivation:

$$\begin{aligned}\sec \theta &= 2 \\ \implies \cos \theta &= 1/2 \\ \implies \theta &= 60^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

4. **4.** The value of $\frac{\tan 30^\circ}{\cot 60^\circ}$ is:

Answer: (A)

Solution:

1. Substitute values: $\tan 30^\circ = 1/\sqrt{3}$ and $\cot 60^\circ = 1/\sqrt{3}$

Derivation:

$$\frac{1/\sqrt{3}}{1/\sqrt{3}} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

5. 5. Which of the following is equal to $\cos \theta$?

Answer: (D)

Solution:

1. Use complementary angle identities: $\cos \theta = \sin(90^\circ - \theta)$

Derivation:

$$\cos \theta = \sin(90^\circ - \theta)$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

6. 6. If $3 \tan \theta = 3$, then θ is:

Answer: (B)

Solution:

1. Simplify the equation to $\tan \theta = 1$

Derivation:

$$\begin{aligned} 3 \tan \theta &= 3 \\ \implies \tan \theta &= 1 \\ \implies \theta &= 45^\circ \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

7. 7. The value of $\sin 0^\circ + \cos 0^\circ$ is:

Answer: (C)

Solution:

1. Evaluate $\sin 0^\circ = 0$ and $\cos 0^\circ = 1$

Derivation:

$$0 + 1 = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

8. 8. If $\sin \theta = \cos \theta$, then θ is:

Answer: (B)

Solution:

1. Divide both sides by $\cos \theta$ to get $\tan \theta = 1$

Derivation:

$$\begin{aligned}\sin \theta / \cos \theta &= 1 \\ \implies \tan \theta &= 1 \\ \implies \theta &= 45^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

9. **9.** Assertion (A): For an acute angle θ , if $\sin \theta = \cos \theta$, then $\theta = 45^\circ$. Reason (R): The value of $\sin \theta$ is equal to $\cos \theta$ only at $\theta = 45^\circ$ in the interval $0^\circ \leq \theta \leq 90^\circ$.

Answer: (A)

Solution:

1. Recall the values of trigonometric ratios.
2. $\sin 45^\circ = \frac{1}{\sqrt{2}}$ and $\cos 45^\circ = \frac{1}{\sqrt{2}}$.
3. Since both are equal, the assertion is true and the reason correctly explains it.

Derivation:

$$\begin{aligned}\sin \theta &= \cos \theta \\ \implies \frac{\sin \theta}{\cos \theta} &= 1 \\ \implies \tan \theta &= 1 \\ \implies \theta &= 45^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

10. **10.** Assertion (A): The value of $2 \sin^2 30^\circ - 3 \cos^2 45^\circ + 1$ is 0. Reason (R): The value of $\sin 30^\circ = 1/2$ and $\cos 45^\circ = 1/\sqrt{2}$.

Answer: (D)

Solution:

1. Calculate $2(1/2)^2 - 3(1/\sqrt{2})^2 + 1$.
2. $2(1/4) - 3(1/2) + 1 = 0.5 - 1.5 + 1 = 0$.
3. Wait, the calculation results in 0, so the Assertion is true. Re-evaluating: $0.5 - 1.5 + 1 = 0$. Both are true.

Derivation:

$$2\left(\frac{1}{4}\right) - 3\left(\frac{1}{2}\right) + 1 = \frac{1}{2} - \frac{3}{2} + 1 = -1 + 1 = 0$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

11. **11.** Assertion (A): The value of $\sin \theta$ can be 2 for some angle θ . Reason (R): The range of $\sin \theta$ is $[-1, 1]$.

Answer: (D)

Solution:

1. The range of $\sin \theta$ is $[-1, 1]$, meaning it cannot exceed 1.
2. Assertion is false as $\sin \theta$ cannot be 2.
3. Reason is true.

Derivation:

$$-1 \leq \sin \theta \leq 1$$

Introduction to Trigonometry — Trigonometric Ratios

- 12. 12.** Assertion (A): In a right-angled triangle, if $\tan A = 3/4$, then $\sin A = 3/5$.
Reason (R): $\sin A = \frac{\text{Opposite}}{\text{Hypotenuse}}$ and $\tan A = \frac{\text{Opposite}}{\text{Adjacent}}$.

Answer: (A)

Solution:

1. Using $\tan A = 3/4$, let opposite side = $3k$ and adjacent = $4k$.
2. Hypotenuse = $\sqrt{(3k)^2 + (4k)^2} = 5k$.
3. $\sin A = 3k/5k = 3/5$.
4. Assertion and Reason are both true and linked.

Derivation:

$$\text{Hypotenuse} = \sqrt{3^2 + 4^2} = 5; \sin A = \frac{3}{5}$$

Introduction to Trigonometry — Trigonometric Ratios

- 13. 13.** Assertion (A): The value of $\cos 0^\circ \cdot \cos 1^\circ \cdot \cos 2^\circ \dots \cos 90^\circ$ is 1. Reason (R): $\cos 90^\circ = 0$.

Answer: (D)

Solution:

1. The expression is a product of terms including $\cos 90^\circ$.
2. Since $\cos 90^\circ = 0$, the entire product is 0.
3. Assertion is false, Reason is true.

Derivation:

$$\cos 0^\circ \cdot \dots \cdot 0 = 0$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 14. 14.** If $\tan A = \frac{3}{4}$, find the value of $\sin A \cos A$.

Answer:

$$\frac{12}{25}$$

Solution:

1. Use the definition of tangent in a right triangle: $\tan A = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{3}{4}$.
2. Find the hypotenuse using Pythagoras theorem: $h = \sqrt{3^2 + 4^2} = 5$. Then $\sin A = \frac{3}{5}$ and $\cos A = \frac{4}{5}$.
3. Calculate the product $\sin A \cos A = \frac{3}{5} \times \frac{4}{5} = \frac{12}{25}$.

Derivation:

$$\begin{aligned}\tan A &= \frac{3}{4} \\ \implies \text{Opp} &= 3k, \text{Adj} = 4k \\ \text{Hyp} &= \sqrt{(3k)^2 + (4k)^2} = 5k \\ \sin A &= \frac{3}{5}, \cos A = \frac{4}{5} \\ \sin A \cos A &= \frac{3}{5} \times \frac{4}{5} = \frac{12}{25}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

- 15. 15.** Evaluate: $\frac{\tan 65^\circ}{\cot 25^\circ} + \sin^2 30^\circ$.

Answer:

$$1.25$$

Solution:

1. Use the complementary angle identity $\tan(90^\circ - \theta) = \cot \theta$ to simplify the first term.
2. Substitute the value of $\sin 30^\circ = \frac{1}{2}$ and compute the sum.

Derivation:

$$\begin{aligned}\frac{\tan(90^\circ - 25^\circ)}{\cot 25^\circ} + \left(\frac{1}{2}\right)^2 \\ = \frac{\cot 25^\circ}{\cot 25^\circ} + \frac{1}{4} \\ = 1 + 0.25 = 1.25\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

- 16. 16.** If $4 \tan \theta = 3$, evaluate $\frac{4 \sin \theta - \cos \theta}{4 \sin \theta + \cos \theta}$.

Answer:

$$\frac{1}{2}$$

Solution:

1. Divide numerator and denominator by $\cos \theta$ to express in terms of $\tan \theta$.
2. Substitute $\tan \theta = \frac{3}{4}$ into the expression.

Derivation:

$$\begin{aligned}\frac{4 \sin \theta / \cos \theta - 1}{4 \sin \theta / \cos \theta + 1} &= \frac{4 \tan \theta - 1}{4 \tan \theta + 1} \\ &= \frac{4(3/4) - 1}{4(3/4) + 1} = \frac{3 - 1}{3 + 1} = \frac{2}{4} = \frac{1}{2}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

- 17. 17.** If $\cos(A + B) = 0$ and $\sin(A - B) = \frac{1}{2}$, find the values of A and B , where $0^\circ < A + B \leq 90^\circ$ and $A > B$.

Answer:

$$A = 60^\circ, B = 30^\circ$$

Solution:

1. Since $\cos(A + B) = 0$, then $A + B = 90^\circ$.
2. Since $\sin(A - B) = 1/2$, then $A - B = 30^\circ$. Solve the system of linear equations.

Derivation:

$$\begin{aligned}A + B &= 90^\circ \\ A - B &= 30^\circ \\ 2A &= 120^\circ \\ \implies A &= 60^\circ \\ 60^\circ + B &= 90^\circ \\ \implies B &= 30^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 18. 18.** Given $\sec \theta = \frac{13}{5}$, calculate $\cot \theta$.

Answer:

$$\frac{5}{12}$$

Solution:

1. Use $\sec \theta = \frac{\text{Hypotenuse}}{\text{Adjacent}} = \frac{13}{5}$. Find the opposite side using Pythagoras theorem.
2. Calculate $\cot \theta = \frac{\text{Adjacent}}{\text{Opposite}}$.

Derivation:

$$\begin{aligned}\text{Opp} &= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \\ \cot \theta &= \frac{\text{Adjacent}}{\text{Opposite}} = \frac{5}{12}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

19. 19. If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B .

Answer:

$$A = 60^\circ, B = 30^\circ$$

Solution:

1. Use the given trigonometric values to equate the angles.
2. Set up a system of linear equations in terms of A and B .
3. Solve the system using the elimination method.

Derivation:

$$\begin{aligned}\sin(A + B) &= 1 = \sin 90^\circ \\ \implies A + B &= 90^\circ \quad (1)\end{aligned}$$

$$\begin{aligned}\cos(A - B) &= \frac{\sqrt{3}}{2} = \cos 30^\circ \\ \implies A - B &= 30^\circ \quad (2)\end{aligned}$$

$$\begin{aligned}\text{Adding (1) and (2): } 2A &= 120^\circ \\ \implies A &= 60^\circ\end{aligned}$$

$$\begin{aligned}\text{Substituting } A \text{ in (1): } 60^\circ + B &= 90^\circ \\ \implies B &= 30^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

20. 20. Evaluate the following expression: $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

Answer:

$$67/12$$

Solution:

1. Substitute the exact values of the trigonometric ratios from the standard table.
2. Simplify the numerator by squaring and performing multiplication.
3. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$ for the denominator.

Derivation:

$$\begin{aligned}\text{Numerator: } 5\left(\frac{1}{2}\right)^2 + 4\left(\frac{2}{\sqrt{3}}\right)^2 - (1)^2 &= 5\left(\frac{1}{4}\right) + 4\left(\frac{4}{3}\right) - 1 \\ &= \frac{5}{4} + \frac{16}{3} - 1 = \frac{15 + 64 - 12}{12} = \frac{67}{12}\end{aligned}$$

$$\text{Denominator: } \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} + \frac{3}{4} = 1$$

$$\text{Total: } \frac{67/12}{1} = \frac{67}{12}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

21. 21. If $15 \cot A = 8$, find the values of $\sin A$ and $\sec A$.

Answer:

$$\sin A = 15/17, \sec A = 17/8$$

Solution:

1. Determine the ratio $\cot A = \frac{8}{15}$.
2. Use the definition $\cot A = \frac{\text{adjacent}}{\text{opposite}}$ to find the third side (hypotenuse) using Pythagoras theorem.
3. Calculate the required ratios $\sin A$ and $\sec A$.

Derivation:

$$\begin{aligned}\cot A &= \frac{8}{15} = \frac{\text{base}}{\text{perpendicular}} \\ \text{Hypotenuse}^2 &= 8^2 + 15^2 = 64 + 225 = 289 \\ \implies \text{Hypotenuse} &= 17 \\ \sin A &= \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{15}{17} \\ \sec A &= \frac{\text{hypotenuse}}{\text{base}} = \frac{17}{8}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

22. 22. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. Hence, or otherwise, evaluate the expression if $\tan \theta = 2$.

Answer:

$$1 + \sec \theta \operatorname{cosec} \theta = 3.5$$

Solution:

1. Convert all terms into $\sin \theta$ and $\cos \theta$.
2. Simplify the denominators by taking common denominators within the fractions.
3. Combine the fractions by finding a common denominator for the entire expression.
4. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$ to simplify the numerator.
5. Separate the terms to match the required RHS.
6. Substitute $\tan \theta = 2$ into the result.

Derivation:

$$\begin{aligned}& \frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}} = \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)} \\&= \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)} = \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\sin \theta \cos \theta (\sin \theta - \cos \theta)} \\&= \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} = \sec \theta \operatorname{cosec} \theta + 1. \text{ For } \tan \theta = 2, \cot \theta = 0.5, \sec \theta = \sqrt{5}, \operatorname{cosec} \theta = \frac{\sqrt{5}}{2}. \text{ Result } = 1 + \frac{5}{2} = 3.5\end{aligned}$$

- 23. 23.** A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 m. At a point on the plane, the angle of elevation of the bottom of the flagstaff is 30° and that of the top of the flagstaff is 60° . Find the height of the tower.

Answer:

3 m

Solution:

1. Let the height of the tower be h m and the distance from the point to the base of the tower be x m.
2. Form the first equation using $\tan 30^\circ$ for the tower height h .
3. Form the second equation using $\tan 60^\circ$ for the total height $(h + 6)$.
4. Express x in terms of h from the first equation.
5. Substitute x into the second equation and solve for h .

Derivation:

$$\begin{aligned}\tan 30^\circ &= \frac{h}{x} \\ \implies \frac{1}{\sqrt{3}} &= \frac{h}{x} \\ \implies x &= h\sqrt{3} \\ \tan 60^\circ &= \frac{h+6}{x} \\ \implies \sqrt{3} &= \frac{h+6}{h\sqrt{3}} \\ \sqrt{3}(h\sqrt{3}) &= h+6 \\ 3h &= h+6 \\ \implies 2h &= 6 \\ \implies h &= 3\end{aligned}$$

Introduction to Trigonometry — Applications of Trigonometry (Heights and Distances)

- 24. 24.** If $\sin A = \frac{3}{5}$, then the value of $\cos A$ is:

Answer: (C)

Solution:

1. Use identity $\sin^2 A + \cos^2 A = 1$
2. Substitute $\sin A = 3/5$
3. Solve for $\cos A$

Derivation:

$$\cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - (3/5)^2} = \sqrt{1 - 9/25} = \sqrt{16/25} = 4/5$$

Introduction to Trigonometry — Trigonometric Identities

25. 25. The value of $\tan 45^\circ$ is:

Answer: (B)

Solution:

1. Recall the trigonometric table for standard angles

Derivation:

$$\tan 45^\circ = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Standard Angles

26. 26. In a right triangle ABC right-angled at B , if $AB = 3$ cm and $BC = 4$ cm, then $\sin A$ is:

Answer: (A)

Solution:

1. Find hypotenuse AC using Pythagoras theorem
2. Use definition $\sin A = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{BC}{AC}$

Derivation:

$$AC = \sqrt{3^2 + 4^2} = 5; \sin A = 4/5$$

Introduction to Trigonometry — Trigonometric Ratios

27. 27. The value of $\cos^2 30^\circ + \sin^2 30^\circ$ is:

Answer: (C)

Solution:

1. Apply the identity $\sin^2 \theta + \cos^2 \theta = 1$ for any angle θ

Derivation:

$$\cos^2 30^\circ + \sin^2 30^\circ = 1$$

Introduction to Trigonometry — Trigonometric Identities

28. 28. If $\tan \theta = \frac{1}{\sqrt{3}}$, then the value of θ is:

Answer: (B)

Solution:

1. Refer to trigonometric ratio table for tan values

Derivation:

$$\begin{aligned}\tan 30^\circ &= 1/\sqrt{3} \\ \implies \theta &= 30^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Standard Angles

29. 29. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is:

Answer: (C)

Solution:

1. Use complementary angle formula $\cos(90 - A) = \sin A$

Derivation:

$$\frac{\sin 18^\circ}{\cos(90^\circ - 18^\circ)} = \frac{\sin 18^\circ}{\sin 18^\circ} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

30. 30. If $3 \cot A = 4$, then $\tan A$ is:

Answer: (D)

Solution:

1. Use relation $\tan A = 1/\cot A$

Derivation:

$$\begin{aligned}\cot A &= 4/3 \\ \implies \tan A &= 1/(4/3) = 3/4\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

31. 31. The value of $9 \sec^2 A - 9 \tan^2 A$ is:

Answer: (A)

Solution:

1. Factor out 9
2. Use identity $1 + \tan^2 A = \sec^2 A$

Derivation:

$$9(\sec^2 A - \tan^2 A) = 9(1) = 9$$

Introduction to Trigonometry — Trigonometric Identities

32. 32. If $\cos \theta = 0$, then the value of θ is:

Answer: (B)

Solution:

1. Check table for cos values

Derivation:

$$\begin{aligned}\cos 90^\circ &= 0 \\ \implies \theta &= 90^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Standard Angles

33. 33. The value of $\sin 0^\circ + \cos 0^\circ$ is:

Answer: (C)

Solution:

1. Plug in values from table

Derivation:

$$0 + 1 = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Standard Angles

34. 34. Assertion (A): The value of $\sin \theta$ can be $\frac{4}{3}$. Reason (R): The value of $\sin \theta$ is always less than or equal to 1.

Answer: (D)

Solution:

1. Check the range of sine function.
2. Since $-1 \leq \sin \theta \leq 1$, the value $4/3$ (which is 1.33) is impossible.
3. The assertion is false because $4/3 \nless 1$.
4. The reason is a standard trigonometric identity/property.

Derivation:

$$\begin{aligned}\text{Assertion: } \sin \theta &= \frac{4}{3} > 1 \\ \implies \text{False. Reason: } -1 &\leq \sin \theta \leq 1 \\ &\implies \text{True.}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

35. 35. Assertion (A): If $\tan \theta = 1$, then $\theta = 45^\circ$. Reason (R): $\tan \theta = 1$ only for $\theta = 45^\circ$ in the range $0^\circ \leq \theta < 90^\circ$.

Answer: (A)

Solution:

1. Recall the trigonometric table for tan values.
2. The value of tan is 1 at 45 degrees.
3. Since the range is restricted to 0 to 90, 45 is the unique solution.

Derivation:

$$\begin{aligned}\tan 45^\circ &= 1. \text{ In } [0, 90), \tan \theta = 1 \\ &\iff \theta = 45^\circ.\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 36. 36.** Assertion (A): The value of $\cos^2 30^\circ + \sin^2 30^\circ$ is 1. Reason (R): $\sin^2 \theta + \cos^2 \theta = 1$ for all θ .

Answer: (A)

Solution:

1. Identify the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$.
2. Since the angle is the same (30 degrees) for both terms, the identity holds.
3. Reason is the correct mathematical explanation for the Assertion.

Derivation:

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ \implies \sin^2 30^\circ + \cos^2 30^\circ &= 1.\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

- 37. 37.** Assertion (A): As θ increases from 0° to 90° , the value of $\sin \theta$ increases. Reason (R): $\sin 0^\circ = 0$ and $\sin 90^\circ = 1$.

Answer: (A)

Solution:

1. Evaluate $\sin 0 = 0$ and $\sin 90 = 1$.
2. Verify the trend of sine function in the first quadrant.
3. The reason correctly explains why the assertion is true.

Derivation:

$$0 < 0.5 < 0.707 < 0.866 < 1 \text{ for } 0^\circ < 30^\circ < 45^\circ < 60^\circ < 90^\circ.$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 38. 38.** Assertion (A): $\sec^2 \theta - 1 = \tan^2 \theta$. Reason (R): $\sec^2 \theta + \tan^2 \theta = 1$ is a standard identity.

Answer: (C)

Solution:

1. Check the validity of Assertion: $1 + \tan^2 \theta = \sec^2 \theta$ is true, so $\sec^2 \theta - 1 = \tan^2 \theta$ is true.
2. Check the validity of Reason: The standard identity is $1 + \tan^2 \theta = \sec^2 \theta$, not $\sec^2 \theta + \tan^2 \theta = 1$.
3. Conclusion: Assertion is true, Reason is false.

Derivation:

$$\begin{aligned}\text{Identity: } 1 + \tan^2 \theta &= \sec^2 \theta \\ \implies \sec^2 \theta - 1 &= \tan^2 \theta.\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

39. If $\tan A = \frac{4}{3}$, find the value of $\sin A + \cos A$.

Answer:

$$7/5$$

Solution:

1. Use the definition of tangent in a right-angled triangle to find the hypotenuse.
2. Calculate $\sin A$ and $\cos A$ and sum them.

Derivation:

Given $\tan A = \frac{P}{B} = \frac{4}{3}$. Let $P = 4k, B = 3k$. By Pythagoras theorem, $H = \sqrt{(4k)^2 + (3k)^2} = 5k$. $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$. $\sin A + \cos A = \frac{4}{5} + \frac{3}{5} = \frac{7}{5}$.

Introduction to Trigonometry — Trigonometric Ratios

40. Evaluate $\frac{2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ}{1}$.

Answer:

$$2$$

Solution:

1. Substitute the standard trigonometric values for 45, 30, and 60 degrees.
2. Simplify the expression.

Derivation:

$\tan 45^\circ = 1$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$. Expression = $2(1)^2 + (\frac{\sqrt{3}}{2})^2 - (\frac{\sqrt{3}}{2})^2 = 2(1) + 0 = 2$.

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles